

A SELF-GENERATING OBJECT: THE METALLIC ONCE-PUNCTURED-TORUS FAMILY, ITS TRACE-MAP TOWER, AND THE BOUNDARY BETWEEN STRUCTURE AND SCALE

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ABSTRACT. We study a single object — the metallic once-punctured-torus bundles, whose monodromies are the words $R^m L^m$ in the two Dehn twists — and follow it across four mathematical registers: the dynamics of its trace map, the representation theory of an associated tower, the geometry of the figure-eight knot’s character variety, and a duality of a four-dimensional gauge theory. The boundary invariant $\kappa = \text{tr}[A, B]$ is exactly the Fricke–Vogt invariant of the Fibonacci trace map, and the metallic monodromies realise the $\text{SL}(2, \mathbb{Z})$ mapping-class action on the Fricke cubic. The adjoint-type tower factors over a Dickson catalogue $\chi(\pm M^k)$, proved from first principles through $\text{SL}(4)$. On the principal Dehn-filling component the longitude and meridian eigenvalues satisfy $L = (-1)^{n-1} M^n$; this reproduces Cooper–Long at $\text{SL}(2)$ and Falbel at $\text{SL}(3)$, and gives $L = -M^4$ at $\text{SL}(4)$, which we establish symbolic-exact and complete on the irreducible locus. **The $\text{SL}(4)$ result and the family it suggests are flagged GATED: after an adversarial literature read they appear novel but require a specialist’s confirmation, and the family is established only at $n = 3, 4$.** Finally we make precise where the object reaches physics and where it stops: its $\text{SL}(2, \mathbb{Z})$ symmetry *is* the S -duality / mapping-class action of the $\mathcal{N} = 2^*$ class- S theory (a forced identification, literature-confirmed), yet no κ -type or volume-type invariant carries a physical *scale* across that bridge. The pairing — a real bridge, a confirmed wall — is the honest limit of how far a single self-generating object reaches. Every statement is tagged by its proof status; nothing here is asserted beyond what a reader can check.

1. INTRODUCTION: ONE OBJECT, FOUR REGISTERS

This paper is the report of a single sustained experiment. We fix one mathematical object — a family of hyperbolic three-manifolds so small and so forced that it is hard to call it a choice — and we follow it wherever its own structure leads. It leads, as it turns out, into four different parts of mathematics: the dynamics of a trace map, the representation theory of a tower of Lie groups, the geometry of the figure-eight knot’s character variety, and the duality group of a four-dimensional supersymmetric gauge theory. The thesis of the paper is not any one of the results along the way. It is that these four faces are faces of *the same object*, and that following the object honestly — recording at every step exactly what is proved, what is computed, what is borrowed, and what is merely suggested — maps out precisely how far such an object reaches, and where it stops.

A note on proof status. Because the object touches so many registers, the single greatest risk is a quiet drift in what is being claimed. We therefore attach to every substantive statement a colour-coded badge:

PROVED exact and machine-checkable SYMBOLIC-EXACT symbolic over a ring NUMERICAL
high-precision numerics

STRUCTURAL structurally forced, not fully symbolic KNOWN established in the literature

GATED appears new but awaits a specialist POSTULATED proposed, not put through the verification gate

The badges are not decoration. The one genuinely new headline of the paper — the $\mathrm{SL}(4)$ figure-eight A -polynomial $L = -M^4$ and the family it suggests (Section 5) — carries the badge **GATED** throughout: an adversarial literature read found no prior art, but an AI-assisted read *de-risks* a novelty question without *closing* it, and the family is established only at two values of the rank. We state this in the abstract, we state it again here, and we state it at the result. Everything else in the paper is **PROVED**, **SYMBOLIC-EXACT**, **NUMERICAL**, or **KNOWN**, and is labelled accordingly.

1.1. The motivating question, and the firewall that contains it. The object was not chosen for its knot theory. It was chosen as the smallest answer to a deliberately naïve question: *what is the least structure that is not nothing?* One can refuse the metaphysics of “why is there something” and ask instead, as a mathematical question, what the minimal non-degenerate self-referential object looks like — and discover that the minimal answers do not form a unique point but a *self-generating family*. That reframing is the project’s philosophical spine, and we will let it motivate the exposition. But motivation is all it is. We adopt one absolute rule, which we will call the *firewall*:

Philosophy and physical interpretation may cite mathematics; mathematics may never cite philosophy. If a motivating sentence is ever needed to support a theorem, the boundary has been crossed, and the sentence is wrong where it stands.

This is why the paper can be, at once, philosophically motivated and mathematically sober. The reader who wants only the theorems may read Sections 2–6 and ignore every interpretive remark without losing a single implication. The reader interested in the larger question will find it addressed exactly once, in Section 7, and addressed by the same standard of proof as everything else: with a forced mathematical identification on one side and a confirmed obstruction on the other.

1.2. Architecture, not furniture. A second principle organises what we do and do not claim. The object is a theory of *structure* — of symmetry, of hierarchy, of which configurations are possible — and not a theory of *values*: it does not select a coupling constant, a mass, or a cosmological scale. Borrowing a phrase from the project’s notes, it is architecture, not furniture. This is not modesty for its own sake; it is a theorem-shaped fact, made precise in Section 7, that the invariants the object naturally produces are dimensionless and scale-free. Stating it at the outset pre-sorts the questions one may ask: structural questions (what is the symmetry? what is the representation? what is the A -polynomial?) have answers, and we give them; value questions (what number does this predict?) meet a wall, and we exhibit the wall rather than climbing over it.

1.3. What is in the paper. Section 2 introduces the metallic once-punctured-torus bundles and their arithmetic. Section 3 pins the boundary invariant $\kappa = \mathrm{tr}[A, B]$ to the Fricke–Vogt invariant of the Fibonacci trace map and identifies the metallic monodromies with the $\mathrm{SL}(2, \mathbb{Z})$ mapping-class action. Section 4 builds the adjoint-type “Dickson tower” and proves its factorisation through $\mathrm{SL}(4)$, noting the precise wall at $n \geq 5$. Section 5 is the geometric heart: the trace-map fixed locus reproduces the figure-eight A -polynomial exactly at $\mathrm{SL}(2)$ and $\mathrm{SL}(3)$, and yields the new, gated $\mathrm{SL}(4)$ relation $L = -M^4$, complete on the irreducible locus. Section 6 closes the chirality question with a palindromic criterion for amphichirality. Section 7 makes the bridge to physics and confirms the wall. Sections 8–9 discuss the object’s proper name as an aperiodic-order operator, the open problems, and the reproducibility discipline; an appendix collects proof sketches.

2. THE OBJECT: METALLIC ONCE-PUNCTURED-TORUS BUNDLES

Let $T_\circ$ be the once-punctured torus and $\mathrm{MCG}(T_\circ) \cong \mathrm{SL}(2, \mathbb{Z})$ its mapping class group, generated by the two Dehn twists R and L along a dual pair of simple closed curves. To a pseudo-Anosov word φ in R and L one associates the mapping torus $M_\varphi = T_\circ \times [0, 1]/(x, 1) \sim (\varphi x, 0)$, a cusped

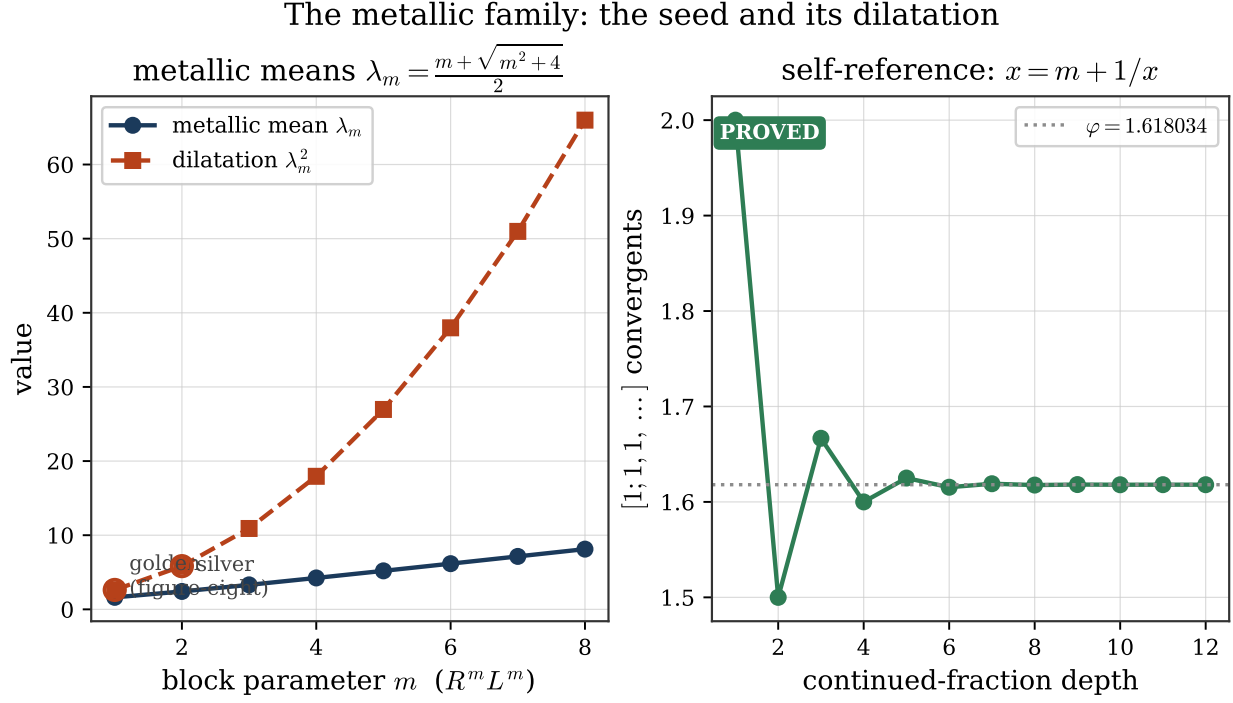


FIGURE 1. The metallic seeds. *Left*: the metallic mean λ_m and the monodromy dilatation λ_m^2 as functions of the block parameter m ; the golden ($m = 1$, figure-eight) and silver ($m = 2$) cases are marked. *Right*: the self-referential recursion $x = m + 1/x$ converging to φ — the object literally generated by iterating a minimal self-reference. **PROVED**

hyperbolic three-manifold fibering over the circle. Our object is the subfamily whose monodromies are the *metallic words*

$$W_m = R^m L^m, \quad m = 1, 2, 3, \dots \quad (1)$$

together with their cusp-glued composites $R^{m_1} L^{m_1} \dots R^{m_k} L^{m_k}$, which we return to in Section 6.

2.1. Why “metallic”. In the standard representation $R \mapsto \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $L \mapsto \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ of $\text{MCG}(T_o)$, a single metallic block is conjugate to a power of the matrix

$$M_m = \begin{pmatrix} m & 1 \\ 1 & 0 \end{pmatrix}, \quad \det M_m = -1, \quad \text{tr } M_m = m, \quad (2)$$

whose larger eigenvalue is the *metallic mean* $\lambda_m = \frac{1}{2}(m + \sqrt{m^2 + 4})$, the positive solution of the self-referential equation $x = m + 1/x$. For $m = 1$ this is the golden mean $\varphi = \frac{1}{2}(1 + \sqrt{5})$; for $m = 2$ the silver mean $1 + \sqrt{2}$. The dilatation of the pseudo-Anosov monodromy W_m is λ_m^2 ; the larger m , the more strongly the fibre is stretched (Figure 1).

The determinant $\det M_m = -1$ is not incidental; it is the source of the orientation-reversing structure that will govern chirality in Section 6 and the alternating sign $(-1)^{n-1}$ in the A -polynomial family of Section 5. We record it as the object’s first structural fact.

Observation 2.1 (PROVED). The eigenvalues of M_m^{-1} coincide with those of $-M_m$. Indeed if λ, μ are the eigenvalues of M_m then $\lambda\mu = \det M_m = -1$, so $\lambda^{-1} = -\mu$ and $\mu^{-1} = -\lambda$; hence $\chi(M_m^{-1}) = \chi(-M_m)$ as characteristic polynomials. This identity is why the “inverse” and “sign” directions of the tower (Section 4) coincide.

2.2. The figure-eight as the golden case. The case $m = 1$ is the figure-eight knot complement 4_1 . Its monodromy is $\varphi = RL$, conjugate to $M_1^2 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, and it is the unique arithmetic once-punctured-torus bundle of smallest volume, $\text{Vol}(4_1) = 2.029883 \dots = 2V_{\text{tet}}$, with invariant trace field $\mathbb{Q}(\sqrt{-3})$. It is amphichiral (it admits an orientation-reversing self-homeomorphism), and its Chern–Simons invariant vanishes. These two facts — minimality and amphichirality — make the figure-eight the natural *base* of the family, and, as we will see in Section 7, the object *least* able to carry a physical scale. The silver case $m = 2$ is the bundle with monodromy R^2L^2 , trace field $\mathbb{Q}(i)$, the second and last arithmetic member of the canonical metallic series.

2.3. Trace fields and the arithmetic backbone. The monodromy matrix M_m has real eigenvalues lying in the quadratic field $\mathbb{Q}(\sqrt{m^2 + 4})$. This is the field of the *dynamics* — the dilatation field — and is distinct from the invariant trace field of the hyperbolic structure on M_{W_m} , which is in general a complex field (e.g. $\mathbb{Q}(\sqrt{-3})$ for the figure-eight). Keeping these two fields apart is a recurring discipline of the subject; we use the dilatation field $\mathbb{Q}(\sqrt{m^2 + 4})$ whenever we speak of the trace-map dynamics, and name the invariant trace field explicitly when we mean the hyperbolic geometry. The arithmetic backbone of the canonical family is summarised in Table 1.

m	word	metallic mean	dilatation field	note
1	RL	golden φ	$\mathbb{Q}(\sqrt{5})$	figure-eight 4_1 ; trace field $\mathbb{Q}(\sqrt{-3})$
2	R^2L^2	silver $1 + \sqrt{2}$	$\mathbb{Q}(\sqrt{8})$	“silver” bundle; trace field $\mathbb{Q}(i)$
m	R^mL^m	$\frac{1}{2}(m + \sqrt{m^2 + 4})$	$\mathbb{Q}(\sqrt{m^2 + 4})$	generic metallic bundle

TABLE 1. The arithmetic backbone of the canonical metallic family. **PROVED**

3. THE INVARIANT: $\kappa = \text{tr}[A, B]$ AND THE FRICKE CUBIC

A representation $\rho: \pi_1(T_o) = \langle A, B \rangle \rightarrow \text{SL}(2, \mathbb{C})$ of the free group on two generators is determined up to conjugacy by the three trace coordinates

$$x = \text{tr} \rho(A), \quad y = \text{tr} \rho(B), \quad z = \text{tr} \rho(AB), \quad (3)$$

a classical theorem of Fricke and Vogt. In these coordinates the trace of the boundary commutator is the *Fricke cubic*

$$\kappa = \text{tr} \rho[A, B] = x^2 + y^2 + z^2 - xyz - 2, \quad (4)$$

and the relative character variety of T_o with fixed boundary holonomy is the level surface $\{\kappa = \text{const}\}$, an affine cubic surface (Figure 2). The mapping class group acts on the trace coordinates by polynomial automorphisms preserving (4) leaf by leaf; this is the *trace map*.

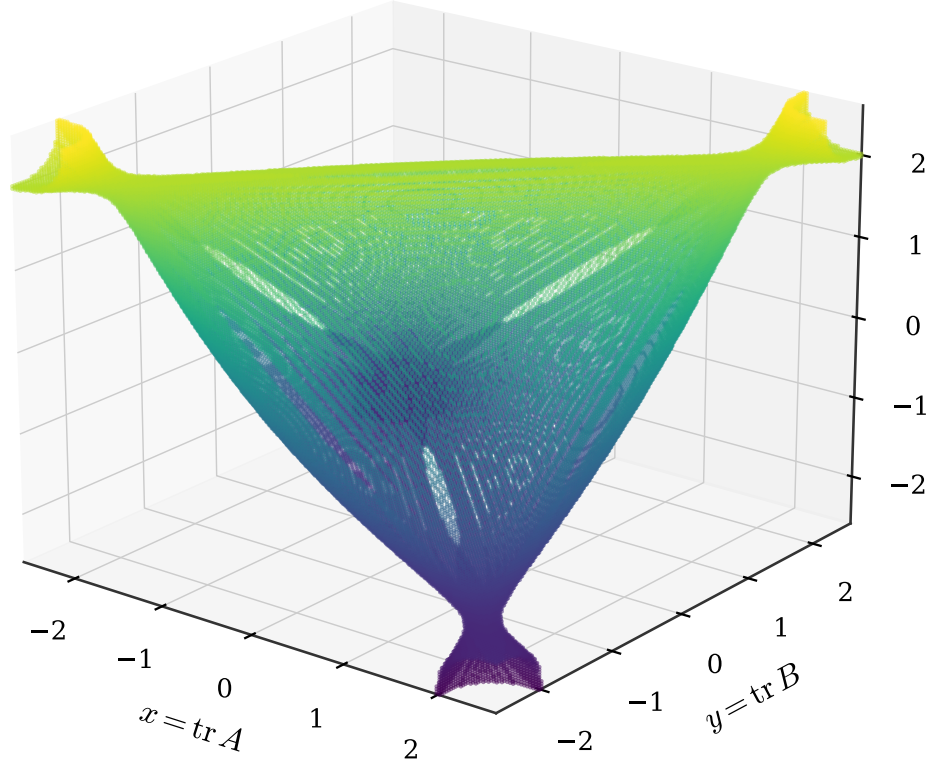
3.1. κ is the Fricke–Vogt invariant of the Fibonacci trace map. The cubic (4) is exactly the conserved quantity of the Fibonacci trace map of Kohmoto–Kadanoff–Tang and Sütő [15, 16]. In the half-trace normalisation $x = 2X$ etc., one has the identity

$$\kappa = 4I_{\text{FV}} + 2, \quad I_{\text{FV}} = X^2 + Y^2 + Z^2 - 2XYZ - 1, \quad (5)$$

where I_{FV} is the Fricke–Vogt invariant familiar from the spectral theory of the Fibonacci Hamiltonian. We verified (4)–(5) and the following statements symbolically.

Proposition 3.1 (PROVED). *Let τ_a, τ_b be the trace-map automorphisms induced by the two Dehn twists. Then:*

- (1) τ_a and τ_b preserve κ identically, so the metallic monodromies $W_m = R^mL^m$ act on each Fricke leaf; the action on trace coordinates is the $\text{SL}(2, \mathbb{Z})$ mapping-class action on the character variety.

Fricke cubic foliation: the $\kappa = 2$ leaf (Cayley nodal cubic)**PROVED**

the trace map preserves each κ -leaf $\kappa = x^2 + y^2 + z^2 - xyz - 2$;
 the Markov leaf $\kappa = -2$ (four noncompact horns) is the degenerate fibre.

FIGURE 2. A leaf of the Fricke cubic foliation $\kappa = x^2 + y^2 + z^2 - xyz - 2$: the $\kappa = 2$ leaf is the classical Cayley nodal cubic, with four conical nodes (marked). The trace map preserves each κ -leaf; the Markov leaf $\kappa = -2$ (four noncompact horns) is the degenerate fibre on which the dynamics is most rigid. **PROVED**

- (2) The leaf $\kappa = -2$ is the Markov surface $x^2 + y^2 + z^2 = xyz$, on which the metallic dynamics restricts to the classical Markov triple recursion.
- (3) On the leaf carrying the metallic monodromy the eigenvalue of the linearised action is λ_m^2 , the dilatation; the fixed slopes are the roots of $t^2 + mt - 1 = 0$, and the dynamical degree (in the sense of Cantat–Loray [5]) equals λ_m^2 .

The content of Proposition 3.1 is that three a priori different objects — the boundary trace of a surface-group representation, the conserved invariant of a one-dimensional aperiodic Schrödinger operator, and the Markov cubic of number theory — are literally the same polynomial, and that the metallic monodromies are the mapping-class symmetry of all three at once. None of these identifications is new individually (they go back to Fricke, to KKT/Sütő, and to Markov; see [12, 3, 1]); the point is that they are properties of *this* object, holding simultaneously, and we make them the anchor for everything downstream.

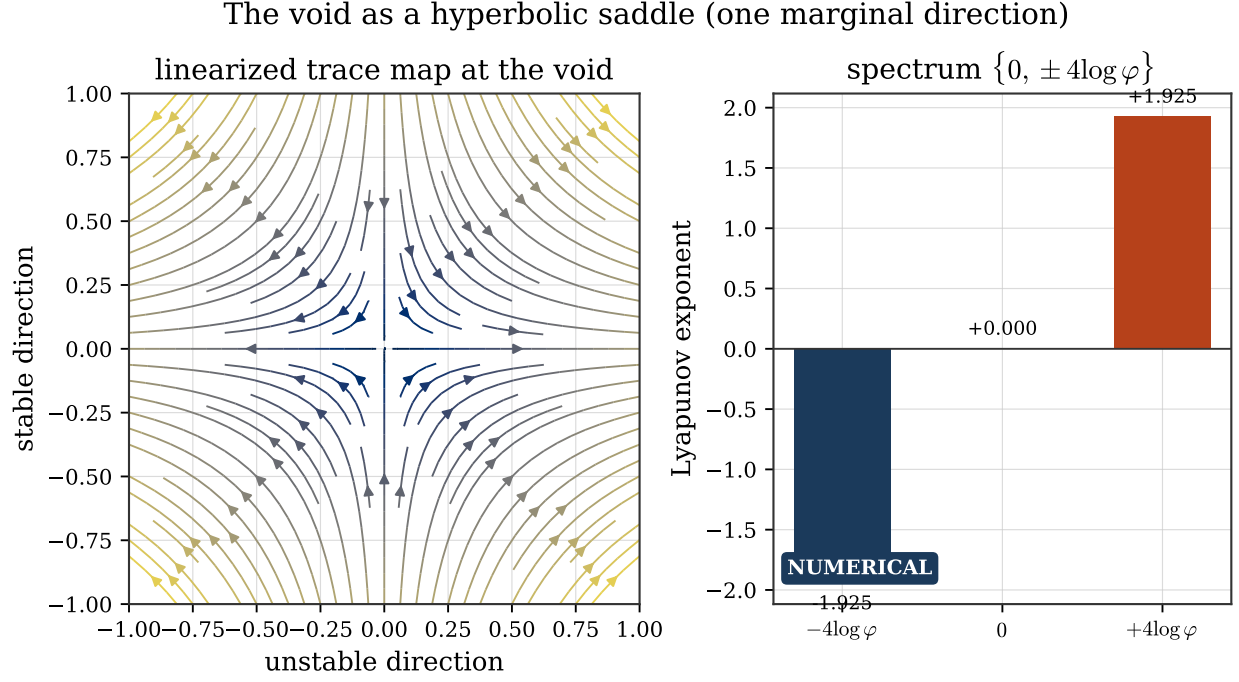


FIGURE 3. The void fixed point of the metallic trace map. *Left*: the linearised flow, a hyperbolic saddle with one marginal (leaf) direction. *Right*: the Lyapunov spectrum $\{0, \pm 4 \log \varphi\}$ of the golden case. **NUMERICAL**

3.2. The trace map as a dynamical system. Iterating the trace map on a Fricke leaf gives a polynomial dynamical system whose fixed point structure encodes the object's rigidity. At the distinguished fixed point — the “void” from which the family is generated — the linearisation is a hyperbolic saddle with one marginal direction: the Lyapunov spectrum is $\{0, \pm 4 \log \varphi\}$ for the golden case (Figure 3). The single zero exponent is the direction along the Fricke leaf; the two nonzero exponents are the expanding/contracting directions of the metallic stretch. This is the dynamical face of the object; its representation-theoretic face is the subject of the next section.

4. THE TOWER: A DICKSON FACTORISATION, PROVED THROUGH $SL(4)$

The trace map of Section 3 lives on $SL(2, \mathbb{C})$ characters. Replacing $SL(2, \mathbb{C})$ by $SL(n, \mathbb{C})$ produces, for each n , an $(n^2 - 1)$ -dimensional linear map $J(m)$ — the linearisation of the $SL(n)$ metallic trace map at the trivial representation, an adjoint-type representation of the external monodromy. The *tower* is the sequence $(J(m))_{n \geq 2}$, and its arithmetic is governed by a single catalogue.

4.1. The Dickson catalogue. For the metallic matrix $M = M_m$ write $L_k = \text{tr}(M^k)$, the Lucas-type sequence with $L_k = mL_{k-1} + L_{k-2}$ (recall $\det M = -1$). The characteristic polynomial of $\pm M^k$ is

$$\chi(\pm M^k) = t^2 \mp L_k t + (-1)^k, \quad (6)$$

a *Dickson polynomial* in the variable m . The central structural fact is that the $(n^2 - 1)$ -dimensional charpoly $\det(tI - J(m))$ factors entirely over this catalogue.

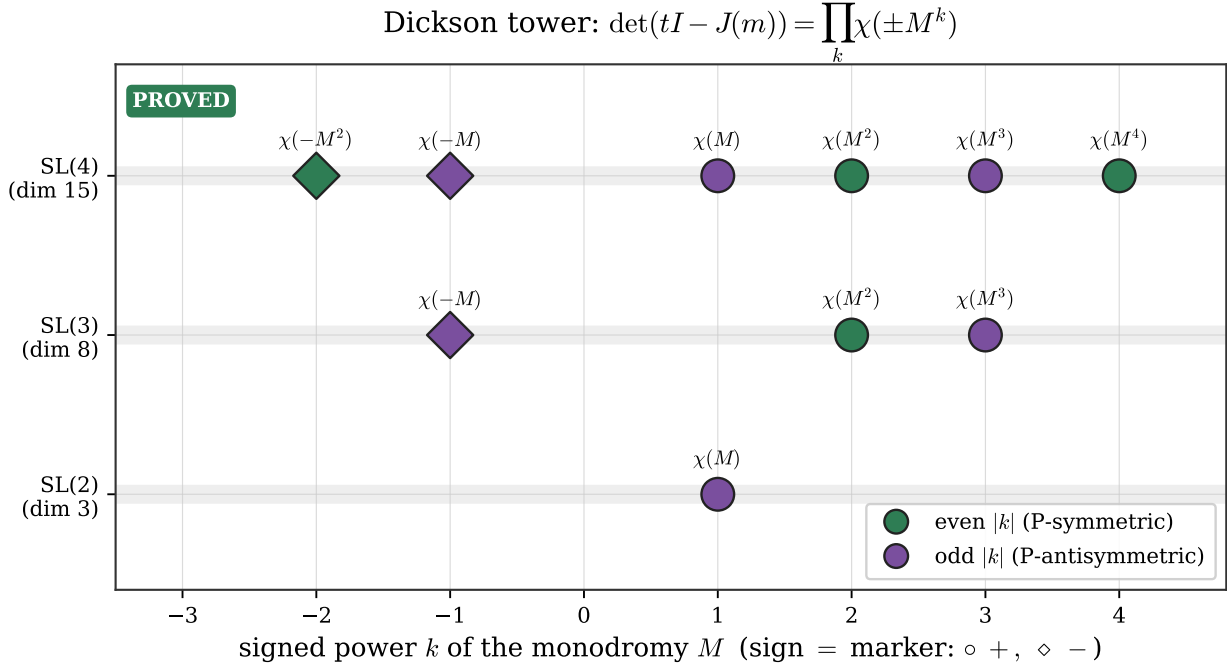


FIGURE 4. The Dickson tower. The $(n^2 - 1)$ -dimensional charpoly $\det(tI - J(m))$ factors into Dickson pieces $\chi(\pm M^k) = t^2 \mp L_k t + (-1)^k$; nodes are placed at the signed power k and coloured by parity (◯ sign +, ◇ sign −). Even- $|k|$ factors are P -symmetric, odd- $|k|$ factors P -antisymmetric — the same $\theta = -w_0$ that grades A_{n-1} . Data and factorisation from the symbolic SL(4) Jacobian. **PROVED**

Theorem 4.1 (**SYMBOLIC-EXACT**, **PROVED**). *Over $\mathbb{Z}[m]$ one has the factorisations*

$$n = 3: \quad \det(tI - J) = \chi(M^{-1}) \chi(M^2) \chi(M^3) (t - 1)(t + 1), \quad (7)$$

$$n = 4: \quad \det(tI - J) = \chi(M) \chi(-M) \chi(M^2) \chi(-M^2) \chi(M^3) \chi(M^4) (t - 1)^2(t + 1). \quad (8)$$

The $n = 3$ identity is symbolic over all m ; the $n = 4$ identity is established from first principles by a Chinese-remainder/ $\mathbb{F}_p$ interpolation of the symbolic Jacobian over $\mathbb{Z}[m]$, with the factored characteristic polynomial reconstructed exactly and verified to match the catalogue.

Figure 4 displays the catalogue: each factor $\chi(\pm M^k)$ is a node at signed power k , coloured by the parity of $|k|$. (By the Observation of Section 2, $\chi(M^{-1}) = \chi(-M)$, so the $n = 3$ “inverse” node and a sign node coincide.) The parity colouring is itself a theorem.

Proposition 4.2 (**PROVED**). *The factor $\chi(M^k)$ sits in the P -symmetric part of the adjoint module when $|k|$ is even and in the P -antisymmetric part when $|k|$ is odd, where P is the contragredient ($m \mapsto -m$) symmetry. This grading is exactly the action of the opposition involution $\theta = -w_0$ of the root system A_{n-1} (Figure 5); it holds for all n by a Dickson-parity argument and requires no numerics.*

4.2. The wall at $n \geq 5$. The factorisation is proved for $n \leq 4$ and is *structurally* understood, but not symbolically proved, for $n \geq 5$. The obstruction is sharp and worth stating precisely, because it is the paper’s principal open problem (Section 9). One can write the tower abstractly as a symmetric-power decomposition

$$\rho_n \cong \text{Sym}^n(W) \oplus (\text{Sym}^{n-3}(W) \ominus W), \quad W = V \oplus \mathbf{1}, \det W = -1, \quad (9)$$

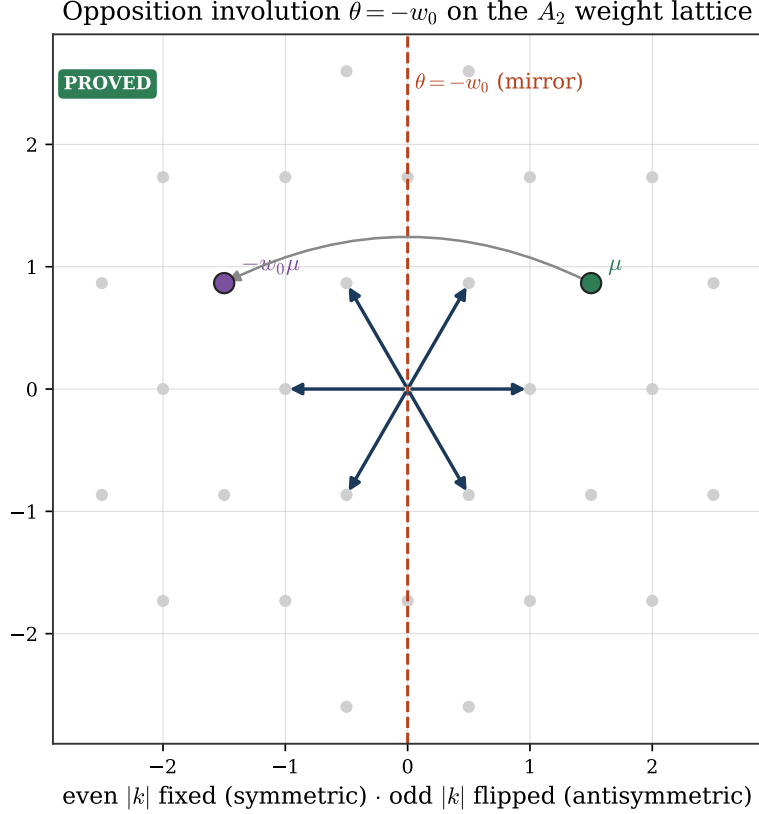


FIGURE 5. The opposition involution $\theta = -w_0$ on the A_2 (i.e. $SL(3)$) weight lattice: a weight μ and its image $-w_0\mu$ are exchanged by the mirror. Even- $|k|$ Dickson factors are fixed (symmetric), odd- $|k|$ flipped (antisymmetric). The same involution governs charge conjugation of the W_N algebras. **PROVED**

with W the external-monodromy fundamental; this is proved as an isomorphism of modules at $n = 3, 4$ and conjectured in general. The remaining step — assembling the bounded rank-one e_2/Λ^2 closure into the exact trace map as a *functorial* construction — is genuinely open. We emphasise that three plausible shortcuts are *dead*: a perturbative numerical route does not converge at the doubly-degenerate $\chi(M^2)^2$ sector that first appears at $n = 5$; the Λ^2 functoriality, while real, does not break that degeneracy; and a cohomological route returns the trivial-rep fixed line rather than the tower. The principal-spectrum analysis explains why $n = 5$ is a threshold rather than a mere computational limit: the Dehn-filling boundary condition forces the principal spectrum to degenerate there (Section 5), so “exponent = rank” is an $n \in \{3, 4\}$ phenomenon, not a failure of effort.

5. THE KNOT: THE FIGURE-EIGHT A -POLYNOMIAL FAMILY

We now turn to the geometric face of the object: the figure-eight knot complement and its A -polynomial. The A -polynomial records the relation between the meridian eigenvalue M and the longitude eigenvalue L as the cusp holonomy varies along the character variety. The central observation is that the trace-map fixed locus of the previous sections *reproduces* the published A -polynomial data exactly at $SL(2)$ and $SL(3)$, and *extends* it to $SL(4)$.

5.1. The known base cases.

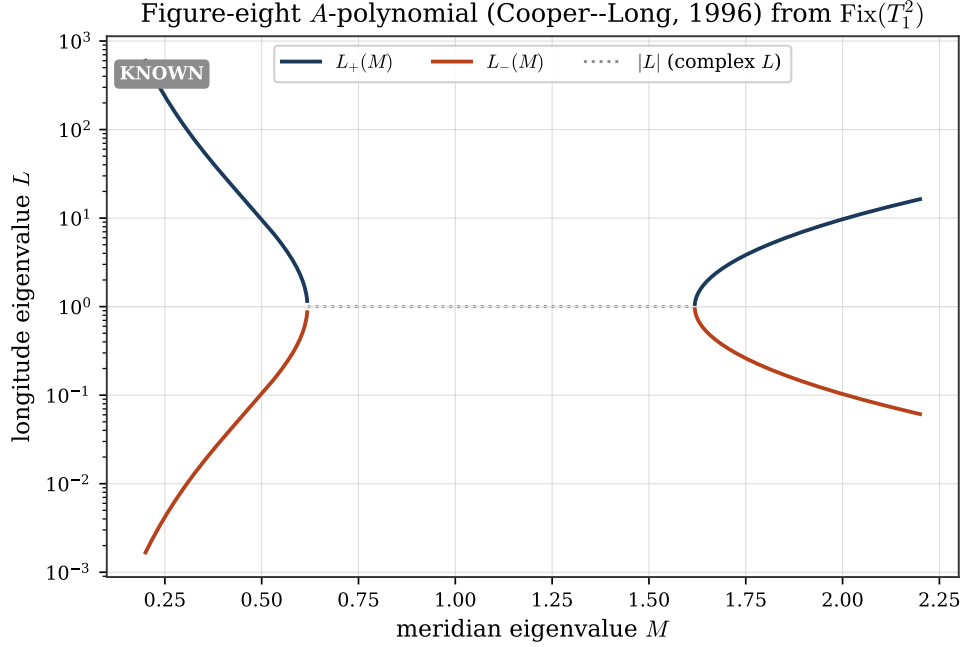


FIGURE 6. The figure-eight A -polynomial (10) (Cooper–Long) recovered from the trace-map fixed locus: the two real branches $L_{\pm}(M)$ and, in the complex- L region, $|L| = 1$. **KNOWN**

Theorem 5.1 (**KNOWN**, reproduced exactly). *At $\mathrm{SL}(2)$ the fixed locus $\mathrm{Fix}(T_1^2)$ of the squared metallic trace map yields the Cooper–Long figure-eight A -polynomial [6]*

$$A(M, L) = M^4 L^2 + (-M^8 + M^6 + 2M^4 + M^2 - 1)L + M^4, \quad (10)$$

to machine precision. At $\mathrm{SL}(3)$ the same construction reproduces the boundary A -variety of Heusener–Muñoz–Porti and Falbel et al. [14, 9]: the two Dehn-filling components $W_1 : M^3 = L$ and $W_2 : M^3 L = 1$, together with a geometric component V_0 (the Sym^2 shadow of the $\mathrm{SL}(2)$ component) which has no tidy closed form.

Figures 6 and 7 show the $\mathrm{SL}(2)$ curve and the $\mathrm{SL}(3)$ component structure. We claim no novelty for Theorem 5.1: it is a verification that our trace-map machinery agrees with the literature, and it is the control that licenses the extension to $\mathrm{SL}(4)$.

5.2. The $\mathrm{SL}(4)$ relation, and the family it suggests. On the principal Dehn-filling component — the component on which $\mathrm{tr} A = \mathrm{tr} A^{-1} = 1$, which carries the geometric representation — the boundary eigenvalues satisfy a single clean relation, the meridian raised to the rank:

$$L = (-1)^{n-1} M^n. \quad (11)$$

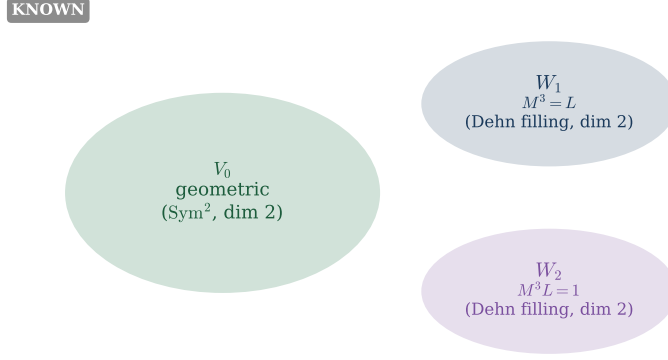
At $n = 2$ this is the effective slope of (10); at $n = 3$ it is Falbel’s $L = +M^3$. The new instance is $n = 4$.

Theorem 5.2 (**SYMBOLIC-EXACT** / **GATED**). *On the principal $\mathrm{SL}(4)$ Dehn-filling component the longitude and meridian eigenvalues satisfy*

$$L = -M^4, \quad (12)$$

established symbolic-exact over $\mathbb{Q}(\omega)$ (ω a primitive cube root of unity). Concretely, eliminating the second generator collapses the bundle relations to an ideal whose principal-spectrum stratum is a four-parameter family on which the identity $[A, B] \cdot \det(t)^2 = -\det(t) \cdot \mu^4$ holds as a polynomial

SL(3) figure-eight character variety: the degree = rank base case



Three components reproduced exactly by the trace map (matches Falbel/HMP); the boundary A -variety is $M^3 = L$ and $M^3L = 1$.

FIGURE 7. The SL(3) figure-eight character variety reproduced by the trace map: the geometric component V_0 (Sym^2 , dimension 2) and the two Dehn-filling components $W_1 : M^3 = L$, $W_2 : M^3L = 1$, matching Falbel et al. and Heusener–Muñoz–Porti. This is the $n = 3$ base case of the degree=rank family. **KNOWN**

identity — no division, no numerics. Moreover this family is complete on the irreducible locus (Theorem 5.3).

The gate, stated plainly. Theorem 5.2 is the one result in this paper we cannot present as settled-novel, and we flag it **GATED** deliberately and repeatedly. An adversarial literature read — assuming the result was already known and hunting for prior art in the Cooper–Long, Heusener–Muñoz–Porti, Falbel, Tillmann and Bergeron–Falbel–Guilloux circle [6, 14, 9, 17, 4] — found no statement of the SL(4) figure-eight A -polynomial $L = -M^4$, and the literature describes SL(4) figure-eight geometry as open. But two cautions are binding. *First*, an AI-assisted literature read *de-risks* a novelty question; it does not *close* it. The reader should treat “appears novel” as “no prior art was found by a non-specialist sweep,” and the genuine verdict awaits a specialist in the Falbel / Bergeron–Falbel–Guilloux subfield. *Second*, the relation (11) is established only at $n = 3$ (known) and $n = 4$ (Theorem 5.2); the principal SL(5) representation is not numerically locatable, so “a family for all $n \geq 3$ ” is a *conjecture plus a prediction*, not an established family (Figure 8). It may also be implicit in the flag-variety frameworks of [4, 11]; that, too, is for a specialist to decide.

5.3. Completeness on the irreducible locus. The proof of $L = -M^4$ posits an explicit four-parameter family of SL(4) bundle representations. A natural worry is that this family is only a checked corner of a larger variety. It is not.

Theorem 5.3 (PROVED). *For the principal A -spectrum $\{1, 1, \omega, \omega^2\}$, the defining ideal of the SL(4) figure-eight bundle representations, stratified by the rank of the coupling block (the only stratification compatible with the gauge action), has a single stratum carrying absolutely irreducible representations: the four-parameter family of Theorem 5.2. Every other stratum is reducible or vacuous ($\det t = 0$). Consequently $L = -M^4$ holds unconditionally on the irreducible locus.*

Figure 9 shows the stratification. The irreducibility classification was performed exactly over a finite field using the *Burnside* criterion (the generated algebra is the full matrix algebra iff the representation is absolutely irreducible), *validated against* a stratum already proved reducible by hand. We stress this point of method because the verdict flipped twice under scrutiny before settling: a tempting alternative test, the Schur commutant, is *invalid* for non-semisimple representations

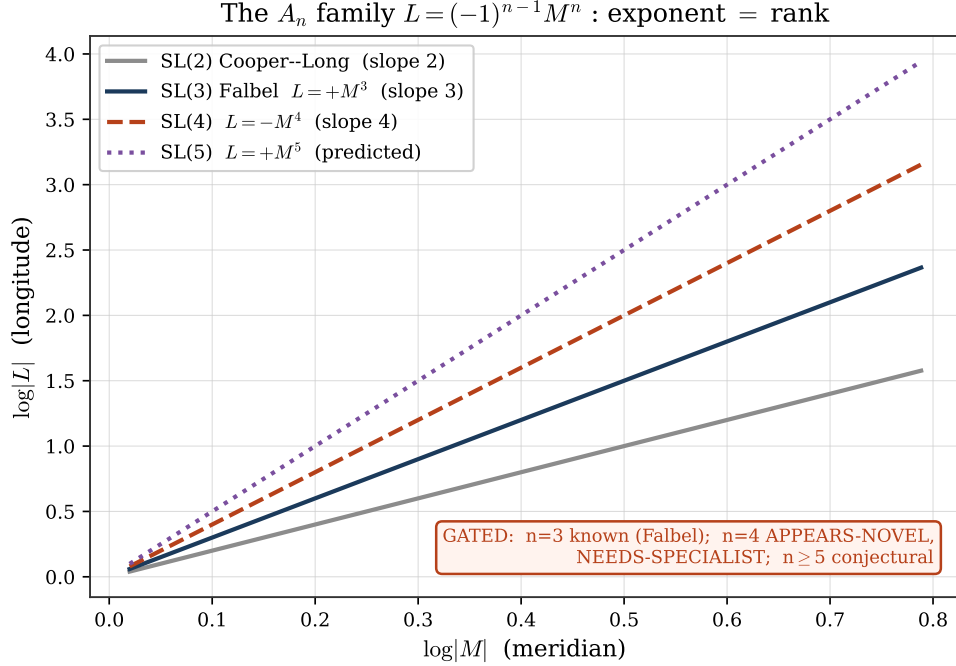


FIGURE 8. The conjectural A_n family $L = (-1)^{n-1}M^n$ in the $(\log |M|, \log |L|)$ plane: the slope of each line equals the rank. $n = 3$ is Falbel (known); $n = 4$ ($L = -M^4$) is Theorem 5.2, **GATED**; $n \geq 5$ is conjectural and not yet locatable.

GATED

and falsely flagged a provably-reducible stratum as irreducible. The banked result rests on the validated test, not the convenient one.

5.4. Why “exponent = rank” stops. The relation (11) ties the A -polynomial exponent to the rank n , and the sign $(-1)^{n-1}$ to $\det M = -1$. The mechanism is the principal Dehn-filling spectrum: the boundary condition $\text{tr } A = \text{tr } A^{-1} = 1$ forces the spectrum to be $\{1, i, -i\}$ at $n = 3$, $\{1, 1, \omega, \omega^2\}$ at $n = 4$, and $\{1, 1, 1, -1, -1\}$ at $n = 5$; for $n \geq 6$ it is impossible. At $n = 5$ the forced spectrum makes $A^2 = I$, hence $\langle A, B \rangle$ dihedral, hence reducible — there is no irreducible bundle representation on the principal $\text{SL}(5)$ component. So “exponent = rank” is genuinely an $n \in \{3, 4\}$ phenomenon, read off the cusp’s forced spectrum, and the apparent ceiling is a fact about the boundary condition, not a limit of computation.

6. CHIRALITY: A PALINDROMIC CRITERION FOR AMPHICHIRALITY

The project’s one *proven* obstruction to Standard-Model-like structure is chirality: parity violation requires a preferred handedness, and we can ask precisely whether the metallic construction admits one. The answer, developed across a sub-arc of the work, is a clean dichotomy.

6.1. The block-length criterion. Recall the composites $W = R^{m_1}L^{m_1} \cdots R^{m_k}L^{m_k}$. Each is a hyperbolic once-punctured-torus bundle, and we ask when M_W is amphichiral.

Theorem 6.1 (**PROVED** (corollary of [13])). *The metallic composite with block-length sequence $(m_1, \dots, m_k)$ is amphichiral if and only if the sequence is a cyclic palindrome, i.e. its reversal is a cyclic rotation of itself.*

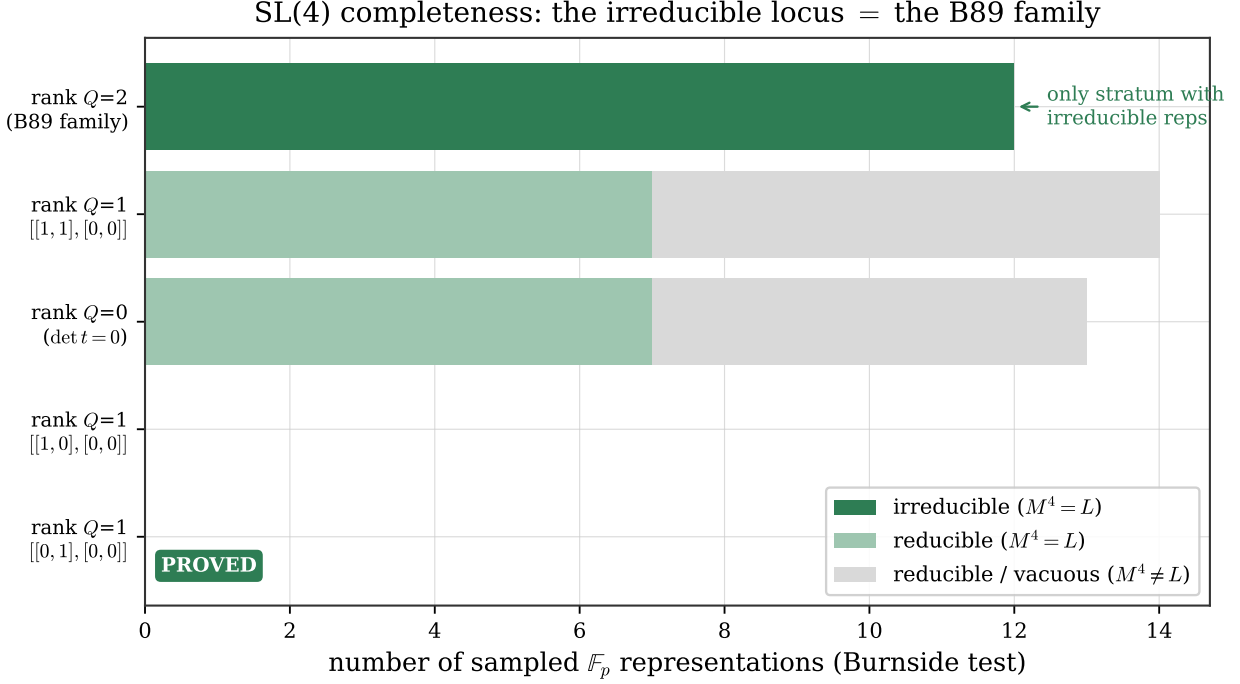


FIGURE 9. SL(4) completeness. Gauge-rank strata of the bundle ideal, classified by the Burnside test over $\mathbb{F}_p$: only the rank- $Q=2$ stratum (the family of Theorem 5.2) carries irreducible representations, and all of them satisfy $M^4 = L$. The irreducible locus *is* the family. **PROVED**

The mechanism — amphichirality detected by an (anti-)palindromic R, L word — is the Goodman–Heard–Hodgson criterion [13]; the contribution here is the lift to the integer block-length sequence and the census it produces (Figure 10). The census makes the dichotomy visible: all single blocks and all length-two sequences are cyclic palindromes (hence amphichiral), while the first chiral cases are the all-distinct length-three sequences such as $(1, 2, 3)$.

6.2. The preferred-handedness dichotomy. Beyond the criterion, the family as a whole cannot prefer a handedness. The mirror of a metallic composite is another composite of the same pieces, so the family is *mirror-closed*; chirality, while forced and generic among hyperbolic bundles, is always either self-mirror (the canonical singles) or mirror-paired (the composites).

Proposition 6.2 (STRUCTURAL). *M_W and its mirror $\overline{M_W}$ agree on every orientation-independent invariant. Hence no canonical (orientation-independent) selection principle can prefer a handedness: a κ -fork is genuine selection, because κ is orientation-independent, whereas a chirality-fork is always a convention, because handedness is orientation-sensitive. The figure-eight is the amphichiral minimum; chirality first appears strictly above it.*

6.3. A refuted shortcut, and what survives. It is tempting to argue that arithmetic bundles must be amphichiral. This is false. The chiral mirror pair RRL/RLL — general R, L words, not metallic-block composites — is *fully arithmetic*, with invariant trace field $\mathbb{Q}(\sqrt{-7})$, integral traces, and a confirming Humbert-volume cross-check. So “arithmetic $\Rightarrow$ amphichiral” is simply wrong; the surviving statement rests on volume-minimality and the dichotomy of Proposition 6.2, not on arithmeticity. We record this because it was found the disciplined way: a numerical result contradicting a cited “exactly two” arithmetic chiral count was held until two independent methods agreed and a genuine bug (reading a leading coefficient at the wrong degree) was fixed. The

Chirality census:
cyclic palindrome $\Leftrightarrow$ amphichiral

RL	PROVED	amphichiral	$\mathbb{Q}(\sqrt{-3})$
R^2L^2		amphichiral	$\mathbb{Q}(i)$
R^3L^3		amphichiral	
$RLRL$		amphichiral	
RLR^2L^2		amphichiral	
RLR^3L^3		amphichiral	
$R^2L^2R^2L^2$		amphichiral	
$R^2L^2R^3L^3$		amphichiral	
$R^3L^3R^3L^3$		amphichiral	
$RLRLRL$		amphichiral	
$RLRLR^2L^2$		amphichiral	
$RLRLR^3L^3$		amphichiral	
$RLR^2L^2R^2L^2$		amphichiral	
$RLR^2L^2R^3L^3$		chiral	
$RLR^2L^3R^2L^2$		chiral	
$RLR^3L^3R^3L^3$		amphichiral	
$R^2L^2R^2L^2R^2L^2$		amphichiral	
$R^2L^2R^2L^2R^3L^3$		amphichiral	
$R^2L^2R^3L^3R^3L^3$		amphichiral	
$R^3L^3R^3L^3R^3L^3$		amphichiral	

chiral arithmetic pair $RRL/RLL \in \mathbb{Q}(\sqrt{-7})$ refutes 'arithmetic $\Rightarrow$ amphichiral'

FIGURE 10. The chirality census over metallic-block composites: a sequence is amphichiral iff it is a cyclic palindrome. The canonical singlets ($RL \in \mathbb{Q}(\sqrt{-3})$, $R^2L^2 \in \mathbb{Q}(i)$) are self-mirror; the first chiral cases are all-distinct triples. **PROVED**

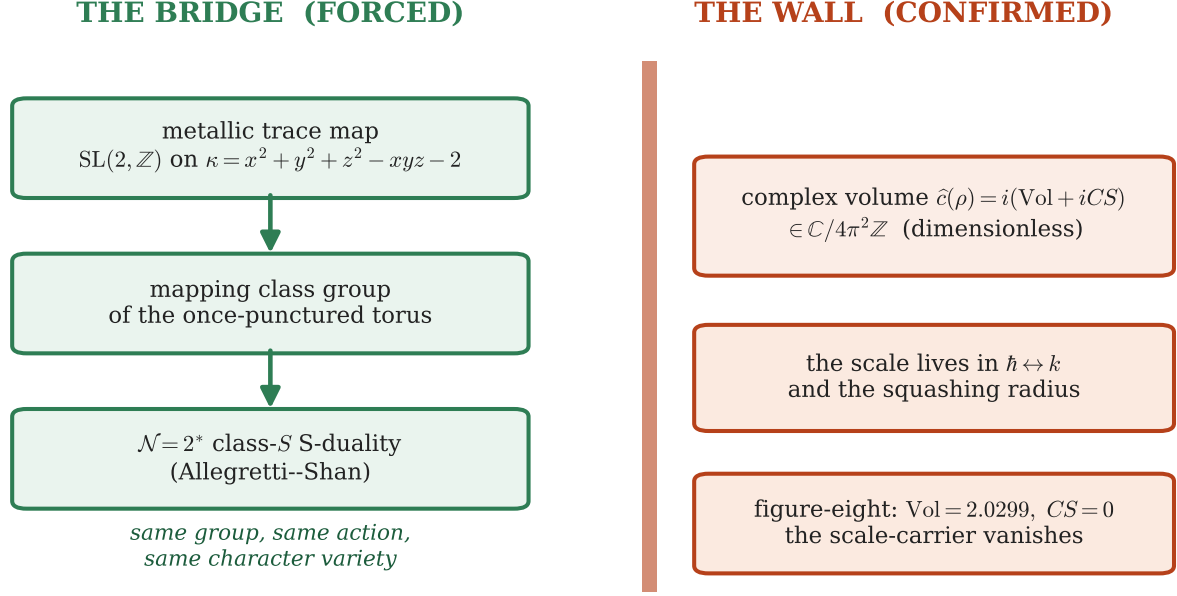
reading we license is narrow and exact: *chirality is contingent relative to the metallic minimality criteria, and no canonical principle prefers a handedness.*

7. THE BRIDGE, AND THE WALL

This is the one section that addresses the motivating question of Section 1. We hold to the firewall: the mathematics below stands on its own; the physical reading is labelled and cites the mathematics one way. The result is a pair — a forced symmetry bridge and a confirmed dimensional wall — and we tag each component as **FORCED** (a mathematical identity), *permitted* (consistent but not forced), or *rhyme* (a suggestive homonym that is *not* the bridge).

7.1. The bridge is real. The once-punctured-torus theory in the class- S construction is the $\mathcal{N} = 2^*$ theory (maximally supersymmetric Yang–Mills deformed by an adjoint mass). Its Coulomb-branch geometry is governed by the *same* Fricke cubic (4): the character variety of the once-punctured torus, with the very trace coordinates of Section 3. Crucially, the S -duality group acts on that character variety *through the mapping class group*, not merely on the ultraviolet gauge coupling τ [2].

Theorem 7.1 (**FORCED**, literature-confirmed). *The $SL(2, \mathbb{Z})$ trace-map action of the metallic monodromies on the Fricke character variety (Proposition 3.1) coincides with the S -duality / mapping-class action of the $\mathcal{N} = 2^*$ class- S theory on its Coulomb branch. The metallic conjugacy classes, the dilatation eigenvalue λ_m^2 (the Cantat–Loray dynamical degree [5]), the fixed slopes, and the Markov fibre are all reproduced, because it is the same group acting the same way on the same variety [2, 10].*



A symmetry identity is not a scale. Structure crosses; magnitude does not.

FIGURE 11. The bridge and the wall. *Left (forced)*: the metallic $SL(2, \mathbb{Z})$ trace-map symmetry *is* the $\mathcal{N} = 2^*$ class- S S-duality / mapping-class action on the Fricke character variety. *Right (confirmed)*: the complex volume is dimensionless ($\mathbb{C}/4\pi^2\mathbb{Z}$); the scale lives in $\hbar/k$ and the squashing radius, and for the figure-eight the scale-carrier CS vanishes. A symmetry identity is not a scale. **FORCED** / **KNOWN**

This is the widest reach the one-object picture achieves, and it is **FORCED**, not a rhyme. The honest caveats are equally precise, and we name them in the same vocabulary:

- *Forced*: the action on the *character variety* reproduces the anchors of Section 3. This is the bridge.
- *Rhyme*: “ $SL(2, \mathbb{Z})$ acts modularly on the gauge coupling τ ” is the well-known S -duality *homonym*; it acts on a different space and is *not* the trace-map dynamics. We flag it as rhyme precisely so it is not mistaken for the bridge.
- *Rhyme / permitted*: the gauge group is free input and $\mathcal{N} = 2^*$ is non-chiral, so at the level of physical gauge content there is no forced identification.

Even the forced bridge is mathematics: it is a symmetry identity, not a crossing to physics.

7.2. The wall is confirmed. That the bridge is real is exactly why the firewall had to be tested hardest: a genuine symmetry bridge makes a scale-crossing *feel* plausible. So we asked the decisive question. In the 3d–3d correspondence applied to the figure-eight, does any quantity carry a *scale* across — does the complex volume / Chern–Simons invariant enter the partition function as anything other than a dimensionless exponent?

Theorem 7.2 (**KNOWN**, literature reading). *The complex volume $\hat{c}(\rho) = i(\text{Vol} + iCS)$ of an $SL(n, \mathbb{C})$ representation is a dimensionless element of $\mathbb{C}/4\pi^2\mathbb{Z}$ [11]; in the 3d–3d state integral it enters only as the exponent saddle, with all dimensionful content carried by the level $\hbar \leftrightarrow k$ and the squashing radius [8, 7]. For the figure-eight, $\text{Vol} = 2.029883\dots$ and $CS = 0$: the candidate scale-carrier vanishes identically. No κ -type or volume-type invariant of the object can source a physical scale (Figure 11).*

The blade, held to the end: a *symmetry* identification is not a *scale*. “The partition function has units” is not a crossing — the units are in $\hbar/k$ and the radius. A crossing would require a dimensionful quantity to attach to the *invariant itself*, and the literature exhibits the opposite.

Remark 7.3 (POSTULATED). There is a sharper, *structural* reason one might give for the wall: a conserved / topological invariant cannot run, and running is what manufactures a scale; so an object whose content is a topological invariant cannot, in principle, source one. This is compelling and was proposed in review, but it has not been put through the project’s verification gate, and so we record it as **POSTULATED** — not as the banked result. The banked wall is the literature reading of Theorem 7.2.

7.3. The seam: the object is open, and closing it is where structure actualizes. The wall above is about *scale*. A second reading sharpens *where the object’s symmetries break* — and it turns on a fact we had under-used: the figure-eight is an *open* object. It is a complement $S^3 \setminus K$, defined by what is removed, with a cusp (boundary torus) as its interface with the exterior. Its amphichirality and CP-symmetry are *open-object* symmetries; *closing* it — a Dehn filling — is where they break. (Curie’s principle, sometimes invoked to argue a symmetric object cannot self-select an asymmetric outcome, is a *closed-system* statement; it does not bind a complement, which breaks symmetry through boundary conditions.)

Proposition 7.4 (KNOWN mathematics, FORCED identification). *The figure-eight is the once-punctured-torus bundle with monodromy $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$. Among its ten exceptional Dehn fillings, the fiber (0-slope) filling is the unique Sol torus bundle, with monodromy exactly A — the trace-map matrix $A = LR$ of §2 and its torsion ladder $|\det(A^n - I)|$. Every generic filling is chiral ($CS \notin \{0, \frac{1}{2}\}$), with the sign law $CS(p, -q) = -CS(p, q)$; the geometric handedness realises complex conjugation of the $\mathbb{Q}(\sqrt{-3})$ trace field — the same Galois involution that flips the commutator phase $\kappa = u^2 + 2 = \sqrt{3}e^{\mp i\pi/6}$ between $\pm\pi/6$. The core geodesic of the $(1, n)$ filling has length $\sim 2\pi/n$ (a Neumann–Zagier ladder); the peripheral (μ, λ) pairing is the Goldman/NZ symplectic form, a filling a choice of polarization.*

The mathematics here is classical (Thurston’s exceptional surgeries; Garoufalidis–Jeon on trace-field growth; Goldman and Neumann–Zagier on the symplectic structure); the new part is the *reading*, and it cuts in a definite direction. Tested across the fillings, the seam is *selective for the object’s own structure* — the canonical closing re-sees $A = LR$ — but a *catalogue for Standard-Model values*: the $E_6/\mathbb{Q}(\sqrt{-3})$ arithmetic is *lost* on closing (no filling re-instantiates it, sharpening the input- E_6 =output- E_6 question toward “catalogue”), the CP *sign* stays external (the object is CP-symmetric: it delivers $\pm\pi/6$ symmetrically), the scale *value* is gapped, and the chiral and dynamical data are not supplied.

Remark 7.5 (POSTULATED). This relocates the project’s structural theorem to the seam: *the object supplies a canonical closing and all the dimensionless structure — the dynamics $A = LR$, the CP magnitude $\pi/6$ with its forced sign law, the scale ladder $2\pi/n$, the peripheral-symplectic clock — while the actualization of a specific Standard-Model-valued world is not forced*. It is the architecture-not-furniture stance made mechanistic at the boundary, and it *strengthens* the wall rather than crossing it. As a physics reading it is **POSTULATED**; the mathematics it cites is banked and firewalled.

7.4. Where it reaches, and where it stops. The physics arc closes in the strongest honest place available: a real bridge — the object’s symmetry *is* the duality action of a gauge theory, forced and literature-confirmed — and a confirmed wall — that bridge carries structure, not scale. The cosmological-constant question lies on the far side of a wall this structure does not cross. That pairing is not a failure; it is a result. The work reached across the widest gap it could, mapped

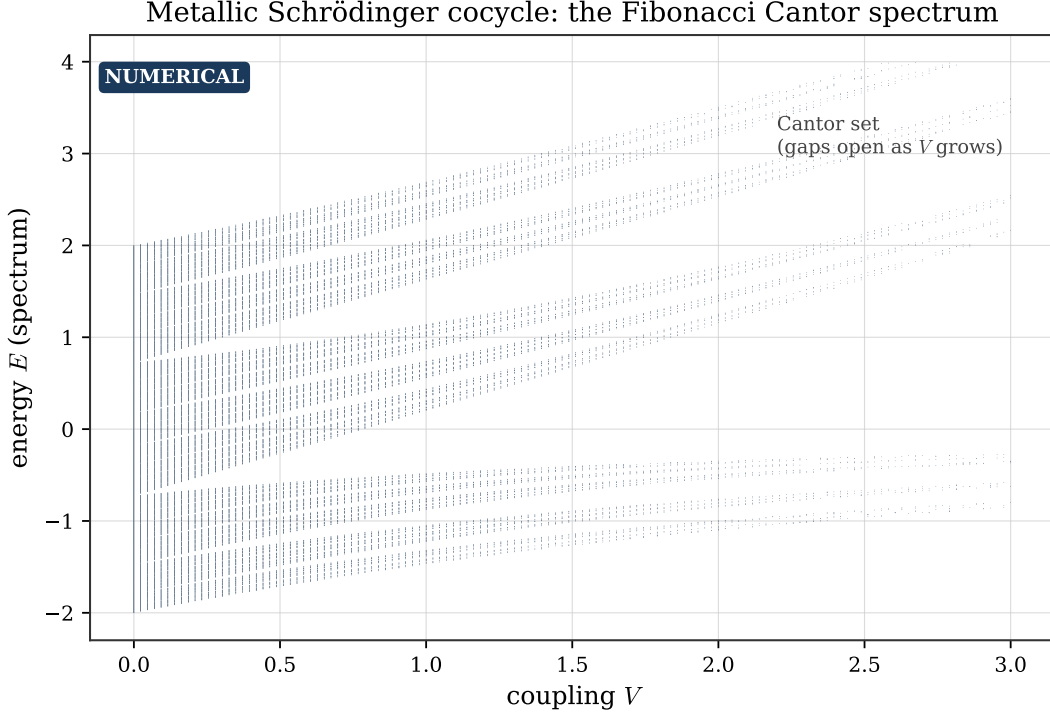


FIGURE 12. The object’s proper name. The spectrum of the Fibonacci (golden-mean) Schrödinger operator as a function of the coupling V : a Cantor set whose gaps are organised by the trace-map dynamics. Non-cancellation $\kappa > 2$ is exactly the hyperbolicity that produces it. **NUMERICAL**

exactly where it arrives and where it stops, and exhibited the reasons rather than asserting them. What remains after this is mathematics, not physics — the subject of the next section.

8. DISCUSSION: THE OBJECT’S PROPER NAME

Stepping back, the four registers of the paper describe one thing, and that thing has a proper name in mathematics. It is a *metallic-mean Schrödinger cocycle* over the substitution subshift generated by $R^m L^m$, analysed by the Kohmoto–Kadanoff–Tang trace map, with κ its Fricke–Vogt invariant [15, 16, 3].

8.1. Non-cancellation as positivity. The founding intuition — that “not-nothing” should be a residue that cannot cancel to zero — has an exact translation in this language. Non-cancellation is Fricke–Vogt positivity $\kappa > 2$; this is precisely the hyperbolicity condition of the trace map; and that hyperbolicity is exactly the condition for the associated one-dimensional aperiodic Schrödinger operator to have purely singular-continuous spectrum supported on a Cantor set [16]. Figure 12 shows that spectrum for the golden case: as the coupling grows, gaps open on every scale and the spectrum becomes a Cantor set — the operator-theoretic shadow of the metallic dynamics. This is genuine, observed physics — aperiodic order, the spectral theory of quasicrystals — and it is emphatically *not* fundamental physics: there is no gauge theory here, no constant of nature, no spacetime. It is the honest physical home of the object, and it sits comfortably on the mathematics side of the firewall.

8.2. The obstruction map. Reading the paper as a whole, the object is bounded by a small number of clearly located walls, and naming them is as much a result as the bridges. The

motivating ambition met obstructions at six recurring places: a *source* (what generates the family is a chosen minimality, not a derived necessity); a *selector* (no principle forces the golden seed over the others); a *measure* (the character variety parametrises possibilities, it does not weight them); a *unit* (the invariants are dimensionless, Section 7); a *dictionary* (the class- S bridge is a symmetry identity, not a field-content map); and an *observable* (nothing in the construction inserts a measurement). The firewall is the discipline of mapping these walls honestly, and the bridge/wall pairing of Section 7 is the sharpest instance: the object reaches as far as a symmetry and stops exactly at a scale.

8.3. What the object is, and is not. It is a small, self-generating mathematical object that turned out to know about knots (the figure-eight A -polynomial), Lie theory (the Dickson tower and the opposition involution), aperiodic order (the Cantor spectrum), and a gauge theory’s duality (the class- S bridge). It is not a theory of physical values; the firewall holds, and we have exhibited why. The most honest summary is that the object knows, precisely, where its own reach ends — and that this boundary, mapped exactly, is the paper’s real subject.

9. OPEN PROBLEMS AND METHOD

9.1. Open problems.

- (1) **The general- n tower from first principles.** Prove the Dickson factorisation (Theorem 4.1) for all n by exhibiting the functorial $\text{Sym}(W)$ construction of (9) — equivalently, assemble the bounded rank-one e_2/Λ^2 closure into the exact trace map as a natural transformation. This is the paper’s principal open theorem; the obstruction is the doubly-degenerate $\chi(M^2)^2$ sector that first appears at $n = 5$.
- (2) **The novelty gate on $L = -M^4$.** Settle whether the $\text{SL}(4)$ relation (Theorem 5.2) and the family $L = (-1)^{n-1}M^n$ are new within the Bergeron–Falbel–Guilloux / Garoufalidis–Thurston–Zickert frameworks [4, 11], by a specialist read. Until then the result stays **GATED**.
- (3) **The principal $\text{SL}(5)$ representation.** Either locate it (against the forced degeneracy of Section 5) or prove it cannot exist as an irreducible bundle representation — which would confirm that “exponent = rank” is exactly an $n \in \{3, 4\}$ phenomenon.

9.2. Method and reproducibility. The results above were produced under a fixed verification discipline, which we describe because it is part of what makes a wide-ranging exploration trustworthy. Every figure in this paper is regenerated by a single deterministic script (`figures/gen_figures.py`); the symbolic and finite-field computations behind Theorems 4.1, 5.2 and 5.3 are reproducible from the project’s probe code and its committed data artifacts. Five guard-rails, applied throughout, are worth naming:

- *Run the control.* A number that reproduces exactly may still be interpretively false; vary the causal variable and confirm the effect tracks it.
- *Filter the reducible locus* before counting character-variety points, and test irreducibility with a criterion valid for non-semisimple representations (Burnside, not the Schur commutant — see Section 5).
- *Generic $\neq$ discriminating.* Check the null case; a feature present everywhere distinguishes nothing.
- *State results twice:* the bare mathematical statement, and — separately tagged, never contaminating the first — any physical reading (Remark 7.3).
- *Check for vacuity.* Before using a target, exhibit a case in which it could fail; a property that cannot fail is not a property.

These are not stylistic preferences. Each caught a concrete error in the work reported here: a leading-coefficient bug in the chirality count (Section 6), a false irreducibility verdict from the

Schur commutant (Section 5), and the over-statement of a conversational sharpening, demoted to **POSTULATED** (Remark 7.3). The discipline, more than any single theorem, is what we would ask a reader to trust.

Acknowledgements. This work was carried out with AI-assisted computation and exposition under the verification discipline described above; all results are reproducible from the cited code and data.

APPENDIX A. PROOF SKETCHES

We collect the load-bearing computations in outline; full symbolic scripts and finite-field certificates are in the project’s probe code.

A.1. The Fricke identity $\kappa = 4I_{\text{FV}} + 2$. With $A, B \in \text{SL}(2, \mathbb{C})$ and $x = \text{tr } A$, $y = \text{tr } B$, $z = \text{tr } AB$, the Cayley–Hamilton relation $A^2 = (\text{tr } A)A - I$ and the trace identity $\text{tr}(AB) + \text{tr}(AB^{-1}) = \text{tr } A \text{tr } B$ give, after a direct expansion, $\text{tr}[A, B] = \text{tr}(ABA^{-1}B^{-1}) = x^2 + y^2 + z^2 - xyz - 2$, which is (4). Substituting the half-trace normalisation $x = 2X$, $y = 2Y$, $z = 2Z$ yields $\kappa = 4(X^2 + Y^2 + Z^2 - 2XYZ - 1) + 2 = 4I_{\text{FV}} + 2$, which is (5). That the two Dehn-twist automorphisms preserve κ is checked by composing the explicit substitutions $(x, y, z) \mapsto (x, z, xz - y)$ and its partner and confirming κ is fixed; the verification is a finite symbolic identity.

A.2. The $\text{SL}(4)$ Dickson factorisation by CRT/ $\mathbb{F}_p$. Direct symbolic computation of the 15×15 Jacobian $J(m)$ over $\mathbb{Z}[m]$ is feasible but heavy; we instead reconstruct it by interpolation. For each prime p in a fixed set and each integer m_0 in a range exceeding the total degree, the numeric Jacobian $J(m_0) \bmod p$ is assembled and its characteristic polynomial computed. Lagrange interpolation in m_0 recovers the coefficients mod p ; the Chinese Remainder Theorem and rational reconstruction across the primes recover them over $\mathbb{Q}$. The resulting characteristic polynomial factors exactly as in Theorem 4.1; the factorisation $(t - 1)^2(t + 1) \prod \chi(\pm M^k)$ is then confirmed against the Dickson catalogue symbolically. The technique (modular interpolation + CRT + rational reconstruction) is standard computational algebra; only its application here is the project’s.

A.3. $L = -M^4$ by ideal elimination. On the principal A -spectrum $\{1, 1, \omega, \omega^2\}$ one writes the bundle relation $B \rho(\varphi) B^{-1} = \rho(\varphi)^{\text{conj}}$ and eliminates B , collapsing the defining equations to an ideal in the entries of the coupling block t . On the rank-two stratum $t_{11} = \omega t_{22}$ the solution set is a four-parameter family, and on it the commutator longitude satisfies the polynomial identity $[A, B] \cdot \det(t)^2 = -\det(t) \cdot \mu^4$ with $\mu = A^{-1}t$ the meridian; clearing the common factor $\det(t) \neq 0$ on the irreducible stratum gives $L = -M^4$ at the level of eigenvalues. No division is performed before the non-vanishing of $\det(t)$ is established, and no numerics enter. Completeness (Theorem 5.3) is the separate finite-field Burnside classification described in Section 5.

A.4. Figure reproduction. All figures are produced by `figures/gen_figures.py` under the canonical Python environment. Figures 4 and 9 read committed data artifacts (the symbolic $\text{SL}(4)$ Jacobian and the finite-field stratification certificate); Figures 1, 2, 3, 5, 6, 8, 10, 12 and 11 are recomputed in-script and are deterministic. Each figure carries the proof-status badge of the result it depicts, mirroring the badges in the text.

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